$$Fe_2O_3.3H_2O \xrightarrow{\Delta} Fe_2O_3\,(s) + 3H_2O\,(g)\uparrow$$

$$Al_2O_3.2H_2O \xrightarrow{\Delta} Al_2O_3\,(s) + 2H_2O\,(g)\uparrow$$

Evaluate yourself 2

2. Magnesite (Magnesium carbonate) is calcined to obtain magnesia, which is used to make refractory bricks. Write the decomposition reaction.

1.3.2 Reduction of metal oxides

Metal oxide can be reduced to crude metal by using a suitable reducing agent like carbon, carbon monoxide, hydrogen, aluminium and other reactive metals such as sodium etc...The choice of reducing agent depends on the nature of the metal. For example, carbon cannot be used as a reducing agent for the reactive metals such as sodium, potassium, aluminium etc...Similarly CO cannot be used to reduce oxides such as ZnO, Al_2O_3. Later in this,we study selection of suitable reducing agents by applying Ellingham diagram.

Smelting

In this method, a flux (a chemical substance that forms an easily fusible slag with gangue) and a reducing agent such as carbon, carbon monoxide (or) aluminium is added to the concentrated ore and the mixture is melted by heating at an elevated temperature (above the melting point of the metal) in a smelting furnace. For example the oxide of iron can be reduced by carbon monoxide as follows.

$$Fe_2O_3\,(s) + 3CO\,(g) \longrightarrow 2Fe\,(s) + 3CO_2\,(g)\uparrow$$

In this extraction, a basic flux, quick lime (CaO) is used. Since the silica gangue present in the ore is acidic in nature, the quick lime combines with it to form calcium silicate (slag).

$$\underset{\text{Flux}}{CaO\,(s)} + \underset{\text{Gangue}}{SiO_2\,(s)} \longrightarrow \underset{\text{Slag}}{CaSiO_3\,(s)}$$

In the extraction of copper from copper pyrites, the concentrated ore is heated in a reverberatory furnace after mixing with silica, an acidic flux. The ferrous oxide formed due to melting is basic in nature and it combines with silica to form ferrous silicate (slag). The remaining metal sulphides Cu_2S and FeS are mutually soluble and form a copper matte.

$$2CuFeS_2\,(s) + O_2\,(g) \longrightarrow 2FeS\,(l) + Cu_2S\,(l) + SO_2\,(g)$$

$$2FeS\,(l) + 3O_2\,(g) \longrightarrow 2FeO\,(l) + 2SO_2\,(g)$$

$$\underset{\text{Gangue}}{FeO\,(s)} + \underset{\text{Flux}}{SiO_2\,(s)} \longrightarrow \underset{\text{Slag}}{FeSiO_3\,(s)}$$

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